INCH-POUND

MIL-DTL-17/90B
19 June 2015
SUPERSEDING
MIL-C-17/90A
9 October 1979

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, COAXIAL, 93 OHMS, M17/90-RG71

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

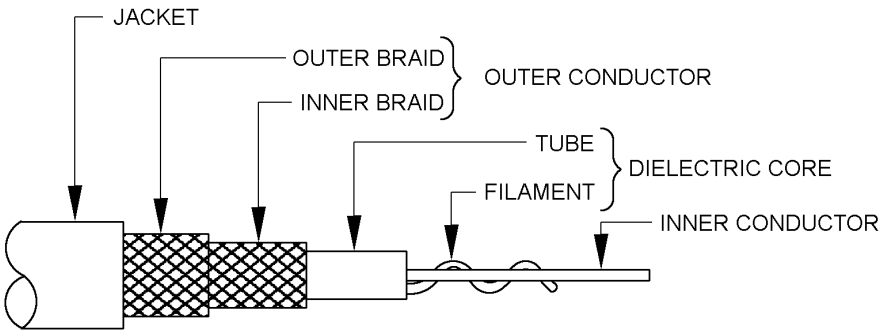


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details	
Inner conductors	Solid, copper-covered, steel wire. Diameter: .0253 ± .0010.	
Dielectric core	Type A-3: Air-spaced polyethylene. A monofilament thread, .035 inch approximate diameter with a lay of 1/2 inch approximate, under an extruded tube. Diameter: .146 inch ± .005. Alternate: A continuous tube, .003 inch thick maximum, under two continuous spiral fins, with a lay of 1-1/4 inches approximate, under an extruded tube.	
Outer conductor	Double braid of copper wire. Diameter: .208 inch, maximum.	
		Alternate
Inner braid	Wire gauge: AWG #34	AWG #36
	Wire coating: Bare	Tinned
	Coverage: 94.3% nominal	95.2% nominal
	Carriers: 16	24
	Ends: 7	6
	Picks/inch: 8.2 ± 10%	12.0 ± 10%
Outer braid	Wire gauge: AWG #36	AWG #36
	Wire coating: Tinned	Tinned
	Coverage: 93.8% nominal	94.6% nominal
	Carriers: 16	24
	Ends: 9	6
	Picks/inch: 10.6 ± 10%	16.0 ± 10%
Jacket	Type IIIa. Diameter: .245 inch ± .005.	

AMSC N/A

FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 84 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +85°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 4.4 ohms per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.
Engineering notes: This cable useful in low temperature applications (see connector series TNC or BNC per MIL-PRF-39012).

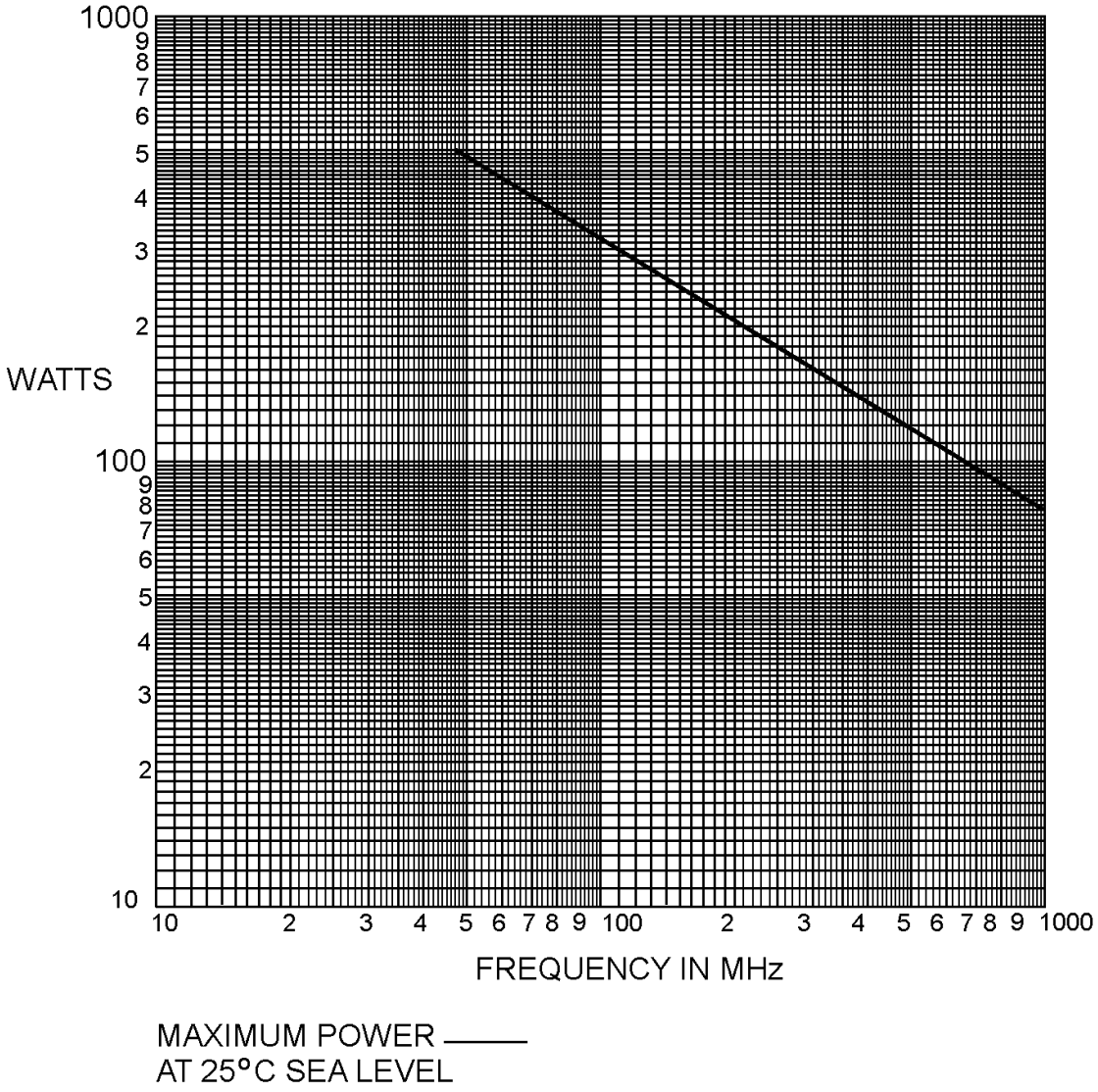
REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adhesion of conductors: Not applicable.
Aging stability: +90° ± 2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -55 ± 2°C.
Dimensional stability: Not applicable.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 5 pounds per 100 feet, maximum.

Electrical:
Test frequency: 50 MHz to 1 GHz.
Spark test: 5,000 V rms, +25%, -0%.
Voltage withstanding: 3,000 Vrms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: Not applicable.
Characteristic impedance: 93 ohms ± 5.
Attenuation: 8.0 dB per 100 feet at 400 MHz.
Structural return loss: Not applicable.
Capacitance: 14.5 pF per foot, maximum.
Capacitance stability: ± 1.5 percent.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

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Power	
MHz	Watts
50	480
100	310
200	200
400	135
1000	77

FIGURE 2. Power rating.

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Part or Identifying Number (PIN): M17/90-RG71.

Supersession data: See table II.

TABLE II. Cross reference of part number.

PIN	Superseded PIN or type designation
M17/90-RG71	RG71B/U

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PFR-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2015-010)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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